Umberto Domenico Ciccia

umbertociccia@icloud.com 085 726 7399 github.com/umbertocicciaa linkedin.com/in/umberto-domenico-ciccia

Experience

Amazon Web Services (AWS), Dublin, Ireland Nov 2025 – Present

Systems Development Engineer

- Eliminated 84 false-positive SEV-2 pages over 6 months by diagnosing a DNS monitoring misconfiguration where same-AZ availability checks failed due to NAT hairpin behavior in a 3-AZ setup
- Built Amazon CloudFront for AWS European Sovereign Cloud end to end, including AWS CDK infrastructure as code, deployment pipelines, service APIs, CloudWatch monitoring and alarms, and on-call operations
- Saved ~4 engineering weeks refactoring 184 Route 53 pipelines (AWS CDK, TypeScript, Ruby) by building a Python AI skill that generated the changes and raised 120 code reviews, raising fresh-artifact pipelines from 80% to 98%
- Delivered Route 53 DNSSEC in AWS European Sovereign Cloud 3 weeks early, unblocking a key customer migration, by building Java APIs, AWS CDK key rotation, CloudWatch alarms, canaries, and console support
- Delivered CIDR-based DNS routing 3 months ahead of schedule, implementing infrastructure, API, and console changes that let customers route DNS queries by client IP range
- Built a Python MCP server letting AI agents query and trigger customer-metadata redaction in logs and metrics through a partner team distributed service (Lambda, SQS with DLQs, SNS, Aurora, DynamoDB)
- Owned on-call incident response for Route 53, AWS WAF, and AWS Shield, writing root-cause analyses, postmortems, and runbooks, and validated peer changes with unit, integration, contract, and canary tests
- Ran sprint planning and agile delivery as the team Scrum Master for 2 quarters

NTT Data, Cosenza, Italy Jan 2025 – Oct 2025

Cloud Software Engineer Consultant

- Built the Amazon EKS cluster and replaced manual Bash deployments with Jenkins CI/CD pipelines automating Terraform provisioning, container image builds to Amazon ECR, and deployments to Amazon EKS
- Added manual approval gates for infrastructure changes and automatic rollback triggered by CloudWatch alarms on API error rate and p50, p90, and p99 latency
- Cut estimated infrastructure costs by 30% versus original designs by building a Quarkus AI agent (Vertex AI SDK) that analyzes multi-cloud architectures on NTT KUMO and recommends optimized configurations
- Developed a Quarkus microservice on NTT KUMO using Crossplane to manage Istio policies across remote Kubernetes clusters, enabling dynamic policy enforcement and reducing manual intervention
- Configured AWS Database Migration Service (DMS) full load and ongoing change replication to migrate application data from on-premises MariaDB to Amazon Aurora
- Built GCP ingestion pipelines processing 10k+ daily records for Coop Italia with Dataflow, Compute Engine, and Cloud Storage, improving data availability and release reliability
- Won an AWS Jam competition representing NTT Data

Vision One, Recanati, Italy July 2024 – Dec 2024

Software Engineer

- Developed an organization-wide on-call scheduling system in .NET and Blazor with rule-based shift assignment, availability validation, and automatic reassignment to resolve conflicts
- Cut time spent on scheduling by 80% by automating scheduling workflows
- Reduced lines of code by 60% through refactoring and added xUnit tests reaching 80% coverage

Caliò Informatica, Cosenza, Italy Jan 2024 – July 2024

Software Engineer

- Cut API costs by 80% and improved performance by 60% in an AI-powered visual workflow builder by eliminating redundant requests and combining calls, validated with before and after profiling

Projects

Unical<3Telegram, unicallovetelegram.github.io/IndiceArgomentiOrale

- Maintained an exam preparation and resource-sharing platform used by 2000+ Università della Calabria students across 8 bachelor and master engineering programs

Skills

Certifications: AWS Solutions Architect – Associate, CKA, GCP Associate Cloud Engineer, Terraform Associate

Cloud and Infrastructure: AWS (CDK, CloudFormation, CloudFront, CloudWatch, Route 53, WAF, Shield, Lambda, SQS, SNS, Aurora, DynamoDB, DMS, EKS, ECR), GCP, Kubernetes, Docker, Terraform, Jenkins, Git, DNS

Languages and Frameworks: Python, Java, TypeScript, C#, Ruby, SQL, Bash, Quarkus, .NET, Blazor

Education

Università della Calabria, Master of Science in Artificial Intelligence and Machine Learning – GPA: 4.0/4.0 2025

Università della Calabria, Bachelor of Science in Computer Engineering – GPA: 4.0/4.0 2024